

2 Warfarin is a drug used to treat people who have a blood clot.

(a) (i) Read through the following passage about warfarin.

Write on the dotted lines the most appropriate word or words to complete the passage.

(3)

Warfarin is used to treat people with blood clots as it lowers the number of clotting factors in the blood.

One clotting factor in blood is prothrombin.

Prothrombin is converted to the enzyme thrombin by

The of thrombin binds to fibrinogen and as a result a mesh of fibres and is formed.

(ii) Which type of drug is warfarin?

(1)

- ☐ **A** an anticoagulant
- ☐ **B** an antihypertensive
- ☐ **C** a platelet inhibitor
- ☐ **D** a statin

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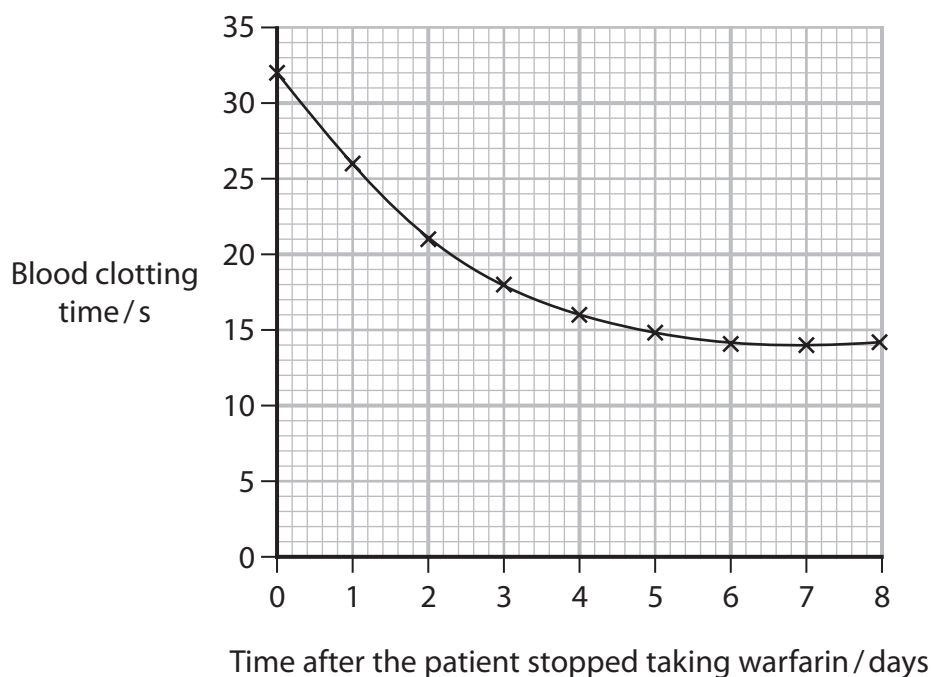
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- (b) The time taken for a blood sample to form a blood clot can be measured. This is called the clotting time.

The graph shows the blood clotting time after a patient has stopped taking warfarin.



Calculate the rate of decrease in the clotting time at two days after stopping taking warfarin.

Use a tangent for your calculation.

(2)

Answer

(Total for Question 2 = 6 marks)

